

# STA347 — Probability

Course Notes

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## Week 1:

### Lecture 1: Set Theory Fundamentals

Sep 8, 2026

“Probability = Set Theory + Measure”

#### 1. Basic Set Operations

Union, Intersect, Complement, ...

$A \cup B$ ,  $A \cap B$ ,  $A^C$ , ...

#### 2. Def 1.1.1 Convergence of Real-Valued Sequences

A sequence  $(x_n)_{n=1}^{\infty} \subseteq \mathbb{R}$  converges to  $x$  if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } n > N \Rightarrow |x_n - x| < \varepsilon.$$

$x_n \rightarrow x$  converge to  $x$ ?

Pick  $\varepsilon$  (how close  $x_n$  is to the target).

For any  $\varepsilon$  picked, can find  $N$ , such that  $n > N$ ,  $|x_n - x| < \varepsilon$ .

e.g.  $x_n = \frac{1}{n} \rightarrow 0$

#### 3. Def 1.1.2 Functional Convergence Pointwisely

A sequence  $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$  converges pointwise to  $f$  (target  $f$ ) if

$$\forall \varepsilon > 0, \forall x \in X, \exists N_x \in \mathbb{N} \text{ s.t. } n > N_x \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

$f_1, \dots, f_n, \dots$  — sequence of functions

$f$  — target function

For any  $\varepsilon$  and  $x$ , can find  $N_x$  s.t.  $n > N_x$ ,  $|f_n(x) - f(x)| < \varepsilon$ .

$f_1(1), f_2(1), \dots, f_n(1), \dots \rightarrow f(1)$  — for every  $x$ , there is a corresponding  $N_x$ .

#### 4. Def 1.1.3 Functional Convergence Uniformly

A sequence  $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$  converges uniformly to  $f$  if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \quad \forall x \in X, n > N \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

That works for all  $x \in X$  — need a common threshold  $N$  for uniform convergence.

#### Key difference:

For pointwise convergence, you need to have  $x$  specified first, then there is a threshold  $N_x$  corresponding to each  $x$ .

For uniform convergence, you need a common threshold  $N$  that works for all  $x \in X$ .

#### Example 1

$$f_n(x) = \frac{x}{n+x}, \quad x \geq 0$$

Then for any fixed value  $x$ , as  $n \rightarrow \infty$ ,  $f_n(x) \rightarrow 0$ :

$$x = 1: f_n(1) = \frac{1}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$x = 2: f_n(2) = \frac{2}{n+2} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\therefore f_n(x) \xrightarrow{\text{pointwise}} 0$$

However,  $f_n$  does not converge uniformly to 0: no  $N$  works for all  $x$ .

No matter how you specify  $N$ , there is always a larger value of  $x$  for which the bound fails.

$$\Rightarrow f_n(x) \not\xrightarrow{\text{uniformly}} 0$$

## Limiting Processes

### Def 1.2.1 The Supremum of a Set

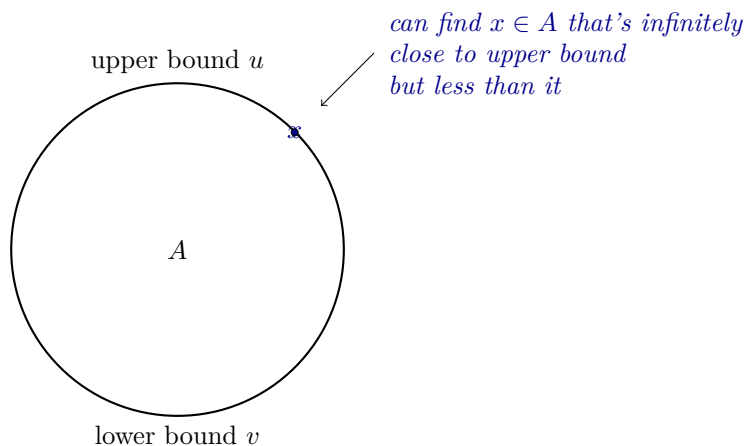
Let  $A \subseteq \mathbb{R}$  be a subset.  $u = \sup A$  iff

$$\forall \varepsilon > 0, \exists x \in A \quad u - x < \varepsilon \quad \text{and} \quad \forall x \in A, x \leq u.$$

### Def 1.2.2 The Infimum of a Set

Let  $A \subseteq \mathbb{R}$  be a subset.  $v = \inf A$  iff

$$\forall \varepsilon > 0, \exists x \in A \quad x - v < \varepsilon \quad \text{and} \quad \forall x \in A, x \geq v.$$



### Example 2

$$A = (0, 1)$$

$$\sup A = 1 \quad \inf A = 0$$

### Distinction of sup/inf vs. max/min:

$\max(A)$  and  $\min(A)$  must be elements of the set  $A$ .

$\sup(A)$  and  $\inf(A)$  may not be elements of  $A$ .

### Example 3

$$A = (0, 1)$$

$\max(A)$  and  $\min(A)$  do not exist.  $\sup(A)$  and  $\inf(A)$  do exist.

$$B = (0, 1]$$

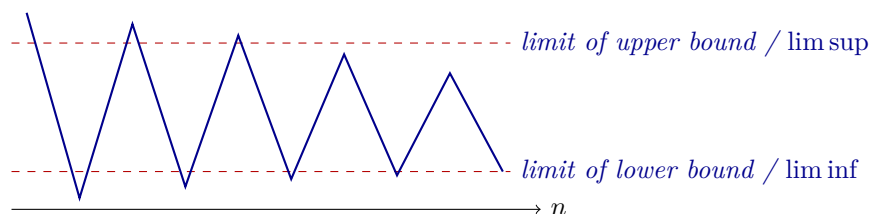
$\min(B)$  does not exist,  $\max(B) = 1$ .  $\sup(B) = 1$ ,  $\inf(B) = 0$ .

### Def 1.2.3 Limsup and Liminf

Let  $(x_n) \subseteq \mathbb{R}$  be a sequence.

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{m \geq n} x_m = \inf_{n \geq 0} \sup_{m \geq n} x_m$$

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{m \geq n} x_m = \sup_{n \geq 0} \inf_{m \geq n} x_m$$



### Example 4

$$x_n = (-1)^n = -1, 1, -1, 1, \dots$$

No limit, diverges:  $\lim_{n \rightarrow \infty} x_n$  DNE.

But  $\sup_{m \geq n} x_m = 1$  and  $\inf_{m \geq n} x_m = -1$

$$\Rightarrow \limsup_{n \rightarrow \infty} x_n = 1, \quad \liminf_{n \rightarrow \infty} x_n = -1$$

*The limsup and liminf are used to describe the long-run behaviour of a sequence, even if the sequence does not converge.*

For a sequence  $x_n$ , if  $\limsup_{n \rightarrow \infty} x_n = \liminf_{n \rightarrow \infty} x_n = x$ , then  $\lim_{n \rightarrow \infty} x_n = x$ .

### Function Limits

### Def 1.13 Left and Right Limit of a Function

For any function  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,

$$f(x-) = \lim_{y \uparrow x} f(y) = L \quad (\text{to the left of } x, \text{ a small neighbourhood})$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in (x - \delta, x), |f(y) - L| < \varepsilon.$$

$$f(x+) = \lim_{y \downarrow x} f(y) = U$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in (x, x + \delta), |f(y) - U| < \varepsilon.$$

If  $f(x-) = f(x+) = C$ , then  $\lim_{y \rightarrow x} f(y) = C$ .

**Example 5**

$$f(x) = \begin{cases} 0 & x \leq 0 \\ 1 & x > 0 \end{cases}$$

$$f(0-) = 0, f(0+) = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

## Lecture 2: Probability Measure

Sep 10, 2026

### Def 2.1 Sample Space

A sample space  $\Omega$  is any non-empty set.

**Example 6** The sample space contains "all the states of the following"

Coin Flipping:  $\Omega : \{H, T\}$

Weather:  $\{\text{hot, cold, mild}\} \times \{\text{wet, dry}\}$

### Def 2.2 Probability Measure (STA347 ver.)

For a sample space  $\Omega$ , we say  $P$  is a probability measure if the following holds:

- i)  $\forall E \subseteq \Omega, 0 \leq P(E) \leq 1$ , that is  $P : \mathcal{P}(\Omega) \rightarrow [0, 1]$
- ii)  $P(\Omega) = 1$
- iii)  $\forall E_1, E_2, \dots \subseteq \Omega$  such that  $E_i \cap E_j = \emptyset \forall i \neq j$

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Remark:

**Example 7** Probabilities of coin flipping

$$\Omega = \{H, T\}$$

$$p(\Omega) = \{\phi, \{H\}, \{T\}, \{H, T\}\}$$

$$P : p(\Omega) \mapsto [0, 1]$$

$$p : \phi \mapsto 0$$

$$\{H\} \mapsto \frac{1}{2}$$

$$\{T\} \mapsto \frac{1}{2}$$

$$\{H, T\} \mapsto 1$$

**Example 8** The uniform/Lebesgue measure Let  $\Omega = [L, U]$  for some  $L < U \in \mathbb{R}$ . Define

*bruh*

Proposition 2.3: The following properties are satisfied by any probability measure  $P$

1.  $\forall E \subseteq \Omega, P(E^C) = 1 - P(E)$

2.  $P(\phi) = 0$
3.  $\forall E, F \subseteq \Omega$ , such that  $E \subseteq F, P(E) \leq P(F)$
4.  $\forall E, F \subseteq \Omega, P(E \cup F) = P(E) + P(F) - P(E \cap F)$
5.  $\forall E, F \subseteq \Omega, P(E \cap F) = P(E) + P(F) - P(E \cup F)$
6.  $\forall E, F \subseteq \Omega, P(F \setminus E) = P(F) - P(E \cap F)$

Proof via Def 1.2 - Probability Axioms

Lemma 2.4

For a sequence  $E_1, \dots, E_n \subseteq \Omega$

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$$

Proof: We define  $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j, F_1 = E_1$

$$\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i \text{ and } F_i \cap F_j = \phi \text{ if } i \neq j$$

from the probability axiom.

$$\begin{aligned} \text{Since } \bigcup_{i=1}^n E_i &= \bigcup_{i=1}^n F_i \\ P\left(\bigcup_{i=1}^n E_i\right) &= P\left(\bigcup_{i=1}^n F_i\right) \\ &= \sum_{i=1}^n P(F_i) \end{aligned}$$

We also know  $F_i \subseteq E_i$  as  $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$

$$\begin{aligned} P(F_i) &\leq P(E_i) \\ P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(F_i) \leq \sum_{i=1}^n P(E_i) \end{aligned}$$

**Week 2:**

## Lecture 3: Set Theory Fundamentals

Sep 8, 2026

### Def Independence 2.7

Events  $E_1, \dots, E_n \subseteq \Omega$  are independent with respect to a probability measure  $P$ , if for each  $\mathcal{I} \subseteq \begin{matrix} [n] \\ \text{any subset of } \{1, \dots, n\} \end{matrix}$

e.g. if  $\mathcal{I} = \{1, 2, 3\}$  a subset of  $\{1, \dots, n\}$

$$p(E_1 \cap E_2 \cap E_3) = p(E_1)p(E_2)p(E_3)$$

**Proposition 2.7.1.** If  $E_1, \dots, E_n$  are independent, then  $E_1^c, \dots, E_n^c$  are also independent. Proof as an exercise.

**Proposition 2.7.2.** An infinite collection of events  $\{E_\alpha : \alpha \in \mathcal{I}\}$  are independent. Then for a finite subset  $\mathcal{J} \subseteq \mathcal{I}$ , the events  $\{E_\beta : \beta \in \mathcal{J}\}$  are independent.

## Random Variables

### Def Random Variables 3.1

#### Def. 3.1 Random Variables

Given a sample space  $\Omega$ , a function  $X : \Omega \rightarrow R$  is a random variable.

#### Example 3.1.1 Coin Flipping

$$\Omega = \{H, T\}$$

$$\downarrow R.V. X : X(H) \rightarrow 1, X(T) \rightarrow 0$$

$$R$$

#### Theorem 3.2 Distribution.

A random variable  $X$  and probability measure  $P$  induce a probability measure  $\mu$  on  $R$  defined by

$$\mu(A) = p(X \in A) = p(\{\omega \in \Omega : X(\omega) \in A\})$$

for any  $A \subseteq R$

$\mu$  : a measure on real number set

$$\mu(\{0, 1\}) = \mu(\{[0, 1]\})$$

$$\mu(A) \text{ (a real number set)} =$$

**Example 3.2.1** Define the following

$$\Omega = \{\text{blue}, \text{red}, \text{green}, \text{yellow}\}$$

$$X : \begin{cases} \text{blue} \rightarrow 1 \\ \text{red} \rightarrow 3 \\ \text{green} \rightarrow 4 \\ \text{yellow} \rightarrow 6 \end{cases}$$

$$A = [1, 3]$$

1.  $\{w : w \in \Omega, X(w) \in A\} = \{\text{blue}, \text{red}\}$
2.  $p(\{\text{blue}, \text{red}\})$

in summary

$$\mu([1, 3]) = \frac{2}{5}$$

$\mu$  is a probability measure for any real number set  $B$ .

proof:

1.  $\mu(B) \in [0, 1]$  is trivial since  $p$  is a probability measure.
2. Since  $X(\omega) \in R$  for all  $\omega \in \Omega$ .

$$\mu(R) = p(\{\omega : \omega \in \Omega, X(\omega) \in R\}) = p(\Omega) = 1$$

3. Consider disjoint sets  $A_1, A_2, \dots \subseteq R$  and define  $E_i = \{\omega : X(\omega) \in A_i\}$   
(e.g.  $E_1 = \{\omega : X(\omega) = 1\}$ )

Since  $X$  is a function, it is impossible for  $X(\omega) = a$  and  $X(\omega) = b$

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} A_i\right) &= p\left(X\omega \in \bigcup_{i=1}^{\infty} A_i\right) \\ &= p\left(\bigcup_{i=1}^{\infty} E_i\right) \quad \text{we know} \\ &= \sum_{i=1}^{\infty} p(E_i) \\ &= \sum_{i=1}^{\infty} \mu(A_i) \end{aligned}$$

### Def. 3.3 The Cumulative Distribution Function

A random variable  $X$  and measure  $p$  generate the cumulative distribution function (CDF) defined by

$$F(x) = \mu((-\infty, x]) = p(X \leq x)$$

where  $p(X \leq x)$  is a short-hand notation for

$$p(\{\omega : \omega \in \Omega, X(\omega) \in (-\infty, x]\})$$

**Theorem 3.3.1.** A distribution function  $F$  for a random variable  $X$  uniquely defines the measure  $\mu$ .

Proof is outside the scope of this class.

**Example 3.3.2 Uniform Distribution**

$P$  is uniform on  $[0, 1]$  and  $X(\omega) = \omega$

For  $x \in [0, 1)$

$$\begin{aligned}
 F(x) &= \mu((-\infty, x]) && \leftarrow \text{by the definition of CDF} \\
 &= p(X \leq x) && \leftarrow \text{by Thm 3.2 (definition of distribution)} \\
 &= p(\{\omega : X(\omega) \in (-\infty, x]\}) \\
 &= p((-\infty, x]) && \leftarrow \text{by } X(\omega) = \omega \\
 &= p([0, x]) && \leftarrow \text{because } P \text{ is a uniform measure} \\
 &= x
 \end{aligned}$$

**Theorem 3.4 Properties of Distribution Function.**

A distribution function satisfies the following properties:

- i) For all  $x \leq y \in R, F(x) \leq F(y)$
- ii)  $\lim_{x \rightarrow -\infty} F(x) = 0$  and  $\lim_{x \rightarrow +\infty} F(x) = 1$
- iii)  $F$  is right-continuous proof:

- i)  $x \leq y \in R$

$$F(x) = \mu((-\infty, x]), F(y) = \mu((-\infty, y])$$

Since  $x \leq y \quad (-\infty, x] \subseteq (-\infty, y]$

$$\mu((-\infty, x]) \leq \mu((-\infty, y])$$

"bigger set, bigger probability"

$$(A \subseteq B, \quad p(A) \leq p(B) \quad / \quad \mu(A) \leq \mu(B))$$

- ii) Define  $A_n = \{X \leq -n\}, \quad n \geq 0$

Then  $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$

$$\begin{aligned}
 \bigcap_{n=1}^{\infty} A_n &= \phi \\
 \lim_{x \rightarrow \infty} F(-n) &= \lim_{x \rightarrow \infty} p(A_n) \\
 &= p\left(\bigcap_{n=1}^{\infty} A_n\right) \\
 &= p(\phi) = 0
 \end{aligned}$$

Define  $B_n = \{X \leq n\},$

then  $B_1 \subseteq B_2 \subseteq \dots$

$$\bigcup_{n=1}^{\infty} B_n = \Omega$$

$$\begin{aligned}
\lim_{x \rightarrow \infty} F(n) &= \lim_{x \rightarrow \infty} p(B_n) \\
&= p\left(\bigcup_{n=1}^{\infty} B_n\right) \\
&= p(\Omega) = 1
\end{aligned}$$

- iii) Fix  $x \in R$  and let  $h_n \downarrow 0$  ( $h_n \geq 0$ ,  $h_n \rightarrow 0$ ,  $h_1 \geq h_2 \geq \dots$ )  
Define  $C_n = \{X \leq x + h_n\}$   
then  $C_1 \supseteq C_2 \supseteq \dots$

$$\lim_{x \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n = \{X \leq x\}$$

Therefore

$$\lim_{x \rightarrow \infty} F(x + h_0) = \lim_{x \rightarrow \infty} p(C_n) = p\left(\bigcap_{n=1}^{\infty} C_n\right) = p(X \leq x) = F(x)$$

Hence

$$\lim_{h \downarrow 0} F(x + h) = F(x)$$

So  $F$  is right-continuous.

### Def. 3.5 The Inverse CDF

The inverse CDF of a r.v.  $X$  is defined by

$$F^{-1}(y) = \sup \{x : F(x) < y\}$$

**Theorem 3.5.1.** If  $F$  satisfies properties i) to iii) of Thm 3.4, then such an  $F$  is the CDF of some r.v.

**Theorem 3.6.** If  $F$  is a CDF, it can only have countably many discontinuities.

## Lecture 4: idk he was super late and stuck at queens park/college turning right

Sep 17, 2026

### Def. 3.7 Density Function

A r.v. has a density function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{if for all } x \in \mathbb{R}$$

$$F(x) = \int_{-\infty}^x f(y) dy$$

we call  $f()$  the density function.

### Def. 3.8 Joint Distribution

We define the joint distribution function of a random vector  $\vec{x} = (x_1, \dots, x_n)$  by

$$F(x_1, \dots, x_n) = p(x_1 \leq x_1, \dots, x_n \leq x_n)$$

using the notation  $p(A, B) \stackrel{\text{def}}{=} p(A \cap B)$

**Theorem 3.8.1.** A joint distribution  $F : \mathbb{R}^n \rightarrow \mathbb{R}$  satisfies the following properties:

i) if  $(x_1, \dots, x_n)$  and  $x'_1, \dots, x'_n$  satisfy  $x_i \leq x'_i$  for all  $i \in [n]$ .

$$F(x_1, \dots, x_n) = F(x'_1, \dots, x'_n)$$

ii)  $\lim_{x_i \rightarrow -\infty} F(x_1, \dots, x_n) = 0$  for all  $i \in \{1, \dots, n\}$  and  $\lim_{x_1 \rightarrow \infty, \dots, x_n \rightarrow \infty} F(x_1, \dots, x_n) = 1$

iii)  $\lim_{h \rightarrow 0^+} F(x_1 + h, \dots, x_n) = \dots = \lim_{h \rightarrow 0^+} F(x_1, \dots, x_n + h)$

$$= F(x_1, \dots, x_n)$$

Proof are analogous to the proof under univariate case.

### Def. 3.9 The Joint Density

A random vector has a joint density function:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \text{if } \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(y_1, \dots, y_n) dy_n \dots dy_1$$

### Theorem 3.10 Independence of Random Variables.

Random variables  $X_1, \dots, X_n$  are independent iff

$$F(X_1, \dots, X_n) = \prod_{i=1}^n F_i(X_i), \quad \forall (X_1, \dots, X_n) \in \mathbb{R}$$

## Expectation

## Lecture 5:

Sep 22, 2026

In STA237:

$$E(X) = \begin{cases} \sum x \cdot p(x) & \text{if } X \text{ discrete} \\ \int x f(x) dx & \text{if } X \text{ continuous} \end{cases}$$

for this course, we are defining probability via "measure theory", so we need to revisit the definition of expectation.

Under the "measure theory" perspective:

$$E(X) = \int x dP \quad \Rightarrow \quad P = \text{probability measure}$$

$$\int x f(x) dx \quad \Rightarrow \quad \int x dP$$

"Reinmann Integral"

### Def. 5.1 Simple Expectation

If  $X$  only takes finitely many values  $x_1, \dots, x_n$ , and  $A_i : \{\omega : X(\omega) = x_i\}$ , define the expectation of  $X$  by:

$$E(X) = \sum_{i=1}^n x_i P(A_i)$$

**Example 5.1.1** For an arbitrary set  $A \subset \Omega$ , let

$$Y(\omega) = I\{\omega \in A\} \quad \text{Then } E(Y) = P(A)$$

$$\begin{aligned} E(Y) &= \sum_{i=0}^1 y_i \cdot P(A_i) \quad Y \text{ only takes two values 0 and 1} \\ &= 1 \cdot P\{\omega \in A\} + 0 \cdot P\{\omega \notin A\} \\ &= P(A) = P(\{\omega : \omega \in A\}) \end{aligned}$$

**Example 5.1.2**  $P$  is uniform measure on  $\Omega = [0, 1]$ , and

$$\begin{aligned} X(\omega) &= \begin{cases} 5 & \omega > \frac{1}{3} \\ 3 & \omega < \frac{1}{3} \end{cases} \\ E(X) &= 5 \cdot P(\omega > \frac{1}{3}) + 3 \cdot P(\omega < \frac{1}{3}) \\ &= 5 \cdot (1 - \frac{1}{3}) + 3 \cdot \frac{1}{3} \\ &= \frac{13}{3} \end{aligned}$$

### Proposition 5.2 Properties of Expectation.

If  $X$  and  $Y$  are simple random variables, the following hold:

- i) If  $x \geq 0$  a.s.,  $E(X) \geq 0$
- ii) For any  $a \in \mathbb{R}$ ,  $E(aX) = a \cdot E(X)$
- iii)  $E(X + Y) = E(X) + E(Y)$

Remark: "a.s." means almost surely. if  $E$  a.s.  $\Rightarrow P(E) = 1$

$x \geq 0$  a.s.  $\Rightarrow P(X \geq 0) = 1 \Leftrightarrow P(\{\omega : X(\omega) \geq 0\}) = 1$

we may have certain  $\omega$ 's s.t.  $X(\omega) < 0$ , but  $P(\{\omega : X(\omega) < 0\}) = 0$

Proof: Both i) and ii) are trivial.

For iii), let  $X$  take values  $x_1, \dots, x_n$  on sets  $A_1, \dots, A_n$  (i.e.  $A_i = \{\omega : X(\omega) = x_i\}$ ) and  $Y$  takes values  $y_1, \dots, y_m$  on sets  $B_1, \dots, B_m$ ,

Observe that there are  $m \cdot n$  events

$$(C_{ij} = \{X = x_i \cap Y = y_j\})$$

Then

$$\begin{aligned} E(X + Y) &= \sum_{i=1}^n \sum_{j=1}^m (x_i + y_j) \cdot P(C_{ij}) \\ &= \sum_{i=1}^n x_i \sum_{j=1}^m P(C_{ij}) + \sum_{j=1}^m y_j \sum_{i=1}^n P(C_{ij}) \\ &= E(X) + E(Y) \end{aligned}$$

**Lemma 5.2.1.** If Proposition 5.2 holds, the following properties also hold:

- iv) If  $X \leq Y$  a.s., then  $E(X) \leq E(Y)$
- v) If  $X = Y$  a.s., then  $E(X) = E(Y)$
- vi)  $|E(X)| \leq E(|X|)$  proof as exercise.

Our logic is, we first define

expectation of simple random variables  
 $\Downarrow$   
 expectation of bounded R.V.'s (approximation)  
 $\Downarrow$   
 expectation of non-negative R.V.'s (truncation)  
 $\Downarrow$   
 expectation of any R.V.'s (symmetric)

### Def. 5.3 Bounded Random Variable

A random variable  $X$  is bounded if there exists  $m < \infty$ , such that  $|x| \leq m$  a.s.

#### Def. 5.3.1 Expectation of Bounded Random Variable

If  $X$  is a bounded R.V., then

$$\begin{aligned} E(X) &= \inf \{E(Z) : Z \text{ simple, } Z \geq X \text{ a.s.}\} \\ &= \sup \{E(Y) : Y \text{ simple, } Y \leq X \text{ a.s.}\} \end{aligned}$$

Recall, when we defined the integral:

the integral is defined as the limit value of sums of areas of tiny rectangles.

When it comes to Reinmann integral, we are no longer cutting  $X$  axis into tiny intervals, but cutting each set into tiny subsets and caculate the sum of  $x$  times the measure of each tiny subset.

$$\begin{aligned} E(X) &= \inf \{E(Z) : Z \text{ is simple and } Z \geq X \text{ a.s.}\} \\ &= \sup \{E(Y) : Y \text{ is simple and } Y \leq X \text{ a.s.}\} \end{aligned}$$

"Squeeze the R.V.  $X$  via two simple R.V.'s  $Y$  and  $Z$ , with one captures the upper bound and the others captures the lower bound".

Proof: Since  $X$  is bounded,  $|x| < m$  a.s.

$$L = \sup \{EY : Y \text{ simple and } Y \leq X \text{ a.s.}\}$$

$$U = \inf \{EZ : Z \text{ simple and } Z \geq X \text{ a.s.}\}$$

$$L \leq U \text{ a.s. } Y \leq X \text{ a.s. and } Z \geq X \text{ a.s.}$$

Our goal is to prove  $L \geq U \Rightarrow L = U \stackrel{def}{=} E(X)$

the way to cut  $[-m, m]$  into tiny intervals.

for any arbitrary  $n$ , we define

$$E_k = \left\{ \omega : \frac{km}{n} \geq X(\omega) \geq \frac{(k-1)m}{n} \right\}$$

we then define two simple R.V.'s

Lower bound:

$$\phi_n(\omega) = \sum_k \frac{(k-1)m}{n} \cdot I(\omega \in E_k)$$

Upper bound:

$$\psi_n(\omega) = \sum_k \frac{km}{n} \cdot I(\omega \in E_k)$$

$$\phi_n(\omega) \leq X(\omega) \leq \psi_n(\omega)$$

$$E\phi_n(\omega) \leq E(X) \leq E\psi_n(\omega)$$

$$E\psi_n(\omega) = E\phi_n(\omega) + \frac{m}{n}$$